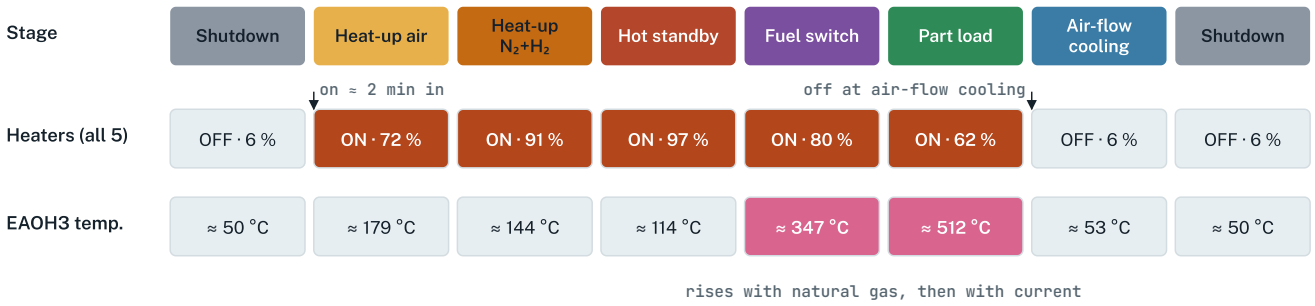
SOFC PLANT · 39 DAY FILES · OCT 2025 – MAR 2026 · DECIDED STAGES

# How the heaters relate to the stages

**Yes, there is a clear relation.** The five gas-line heaters switch on and off with the stages, and their temperatures fall into three steady patterns. They are good enough to **confirm** a stage, but not to **replace** the air, gas and current signals that decide it.

This page uses the eight stages decided on 2 November: **Heat-up air, Heat-up $N_2$+$H_2$, Hot standby, Fuel switch, Part load, Air-flow cooling, Shutdown** and **Trip: purge**. The stack is the fuel cell itself. The heaters are small electric heaters on the gas pipes going into and out of it: ERH on the reformer, EAGH on the fuel line into the stack, EAOH1 and EAOH2 on the exhaust-gas lines to the burner, and EAOH3 on the line that recycles exhaust gas back to the fuel side.

▶ Play a typical run




A typical run from cold to part load and back. Middle row: share of minutes with the heaters on in that stage. Bottom row: median EAOH3 temperature in that stage. The heaters work as one switch: on while the plant heats or holds, off once it cools down.

## Three things the heaters tell us

**93 %**

of minutes where **heaters on** matches an **active stage** (the two heat-ups, hot standby, fuel switch,

**3 groups**

Once the stack is hot, ERH runs 95–196 K **hotter** than it. EAGH, EAOH1 and EAOH2 stay

**+8.6 K per A**

EAOH3 warms as the stack current rises, four times faster than the stack. Above 365 °C it

part load) and **heaters off** matches air-flow cooling, shutdown or trip purge.

Each stage's pattern →

**within ±90 K** of it. EAOH3 stays **far colder**.

See the temperatures →

marks 99 % of part-load minutes.

Current and gas →

## What this means in practice

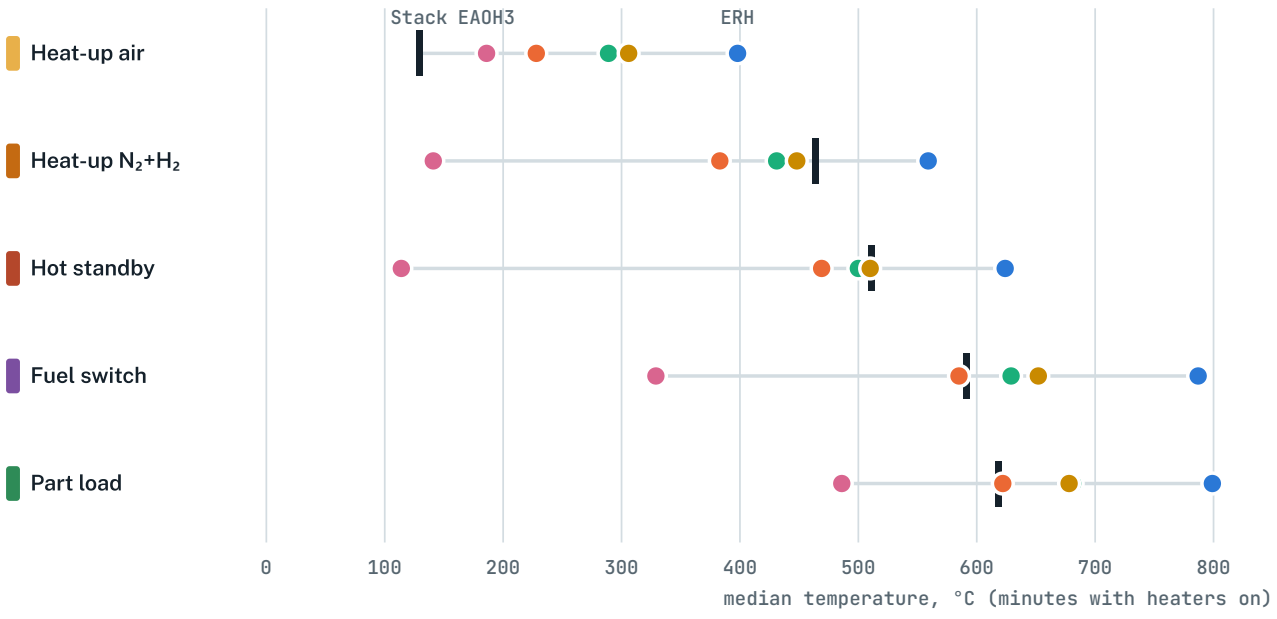
The heaters can act as a second pair of eyes on the stage labels: a quick test that flags minutes where the stage and the heaters disagree. On their own they cannot tell the hot stages apart (Heat-up $N_2$+$H_2$, Hot standby, Fuel switch), and they are off in both Air-flow cooling and Shutdown, so they cannot see the air flow that separates those two. What we can decide →

> **If you remember one thing:** heaters on = plant heating or holding, heaters off = cooling or shut down. EAOH3 hot = natural gas and current flowing.

PAGE 2 · THE HEATER PATTERN OF EACH STAGE

# Each stage has its own heater pattern

In every active stage the same order shows up among the heaters: **ERH hottest, EAOH3 coldest**, and the other three in between. From Heat-up $N_2$+$H_2$ onwards the stack sits among those three, and the overall level simply follows the stack. Only in **Heat-up air** is the stack still colder than every heater: the heaters are warming the gas lines ahead of it.




Median temperature over the minutes when the heaters were on, all 39 day files. The black bar is the stack; dots are the heaters. Hover a mark to read its value.

## Heater minus stack, in kelvin

A positive number means the pipe at that heater is hotter than the stack. One kelvin (K) is the same step as one degree Celsius.

| STAGE | ERH | EAGH | EAOH1 | EAOH2 | EAOH3 |
|---|---|---|---|---|---|
| Heat-up air | +269 | +99 | +160 | +177 | +57 |
| Heat-up $N_2$+$H_2$ | +95 | -81 | -33 | -16 | -323 |

| STAGE | ERH | EAGH | EAOH1 | EAOH2 | EAOH3 |
|---|---|---|---|---|---|
| Hot standby | +113 | -42 | -11 | -1 | -397 |
| Fuel switch | +196 | -6 | +38 | +61 | -262 |
| Part load | +181 | +4 | +63 | +60 | -132 |

- **ERH** runs well above the stack in every stage. It heats the reformer, which needs to be hotter than the stack.
- **EAGH, EAOH1, EAOH2** sit within ±90 K of the stack once it is hot. They mostly show the temperature of the gas passing through them.
- **EAOH3** is 132–397 K below the stack once it is hot, although it draws the most current (8.4 A against 3.2 A for ERH). It starts to close the gap in Fuel switch and most in Part load.

## How often the heaters are on, per stage

| STAGE | HEATERS ON |
|---|---|
| Heat-up air | 72 % |
| Heat-up $N_2$+$H_2$ | 91 % |
| Hot standby | 97 % |
| Fuel switch | 80 % |
| Part load | 62 % |
| Air-flow cooling | 6 % |
| Shutdown | 6 % |
| Trip: purge | 14 % |

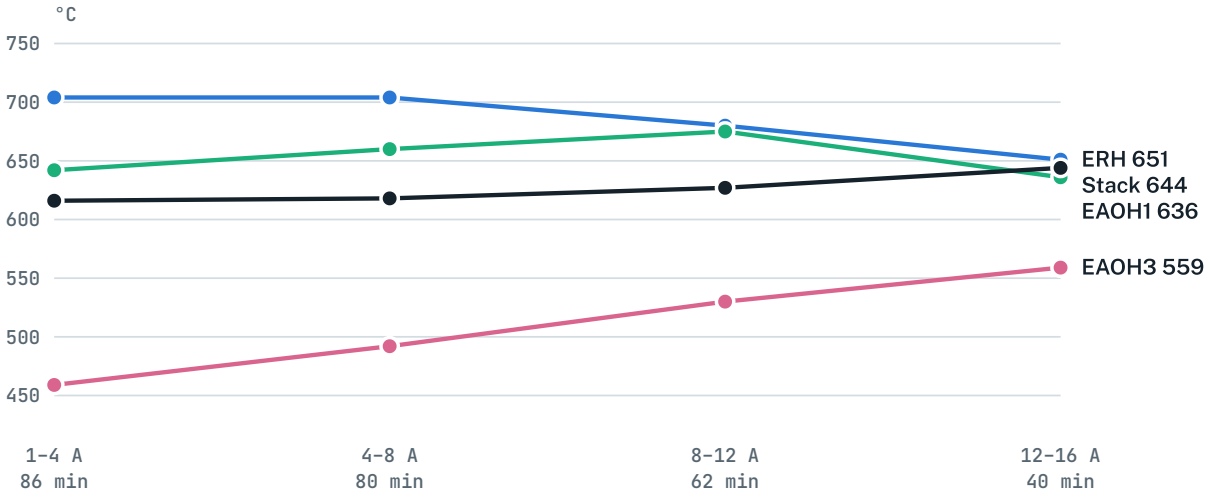
Hot standby and Heat-up $N_2$+$H_2$ are almost always heated. Air-flow cooling, Shutdown and Trip: purge are almost never heated. The exceptions are known cases: in **Shutdown** a heater switched on without air (false start, 632 min); in **Air-flow cooling** the first minutes of air cooling right after a stop (656 min) and a slow cool-down (283 min); in **Heat-up air** the first part of some heat-ups, where the air heater warms the stack before the gas-line heaters join (940 min). In part load only 62 % of minutes have the heaters on: some heaters drop out when the stack's own heat takes over.

> **If you remember one thing:** once the stack is hot the order is always ERH > EAGH / EAOH1 / EAOH2 ≈ stack > EAOH3. The stage changes the level, not the order.

PAGE 3 · CURRENT PRODUCED AND NATURAL GAS

# EAOH3 is the heater that feels the current

When the stack makes power, only **EAOH3** responds clearly: it gets warmer the more current flows. The stack itself warms a little. ERH actually gets cooler.




Part-load minutes (268 in total) grouped by stack current per string. Median temperature in each group. Hover a point to read its value.

## How strongly each one follows the current

| SIGNAL | CHANGE PER EXTRA AMP | LINK TO CURRENT (-1 TO 1) |
|---|---|---|
| EAOH3 | +8.6 K | 0.69 |
| Stack | +2.0 K | 0.36 |
| EAOH1 | +0.3 K | 0.03 |
| ERH | falls (704 → 651 °C) | -0.15 |

The "link" is a correlation: 1 means the two always rise together, 0 means no relation. Only EAOH3 has a strong link (0.69).

## A simple test: EAOH3 above 365 °C

**264 of 268**

**196**

**116 → 470 °C**

part-load minutes have EAOH3 above 365 °C. It almost never misses a load period.

other minutes are also above 365 °C: 89 in **Fuel switch** (natural gas on, no current yet) and 107 in the first minutes after a load ends (Hot standby, Air-flow cooling, Trip: purge, Shutdown).

median EAOH3 with a hot stack, natural gas off → on. The recycle line warms when real fuel gas flows through it.

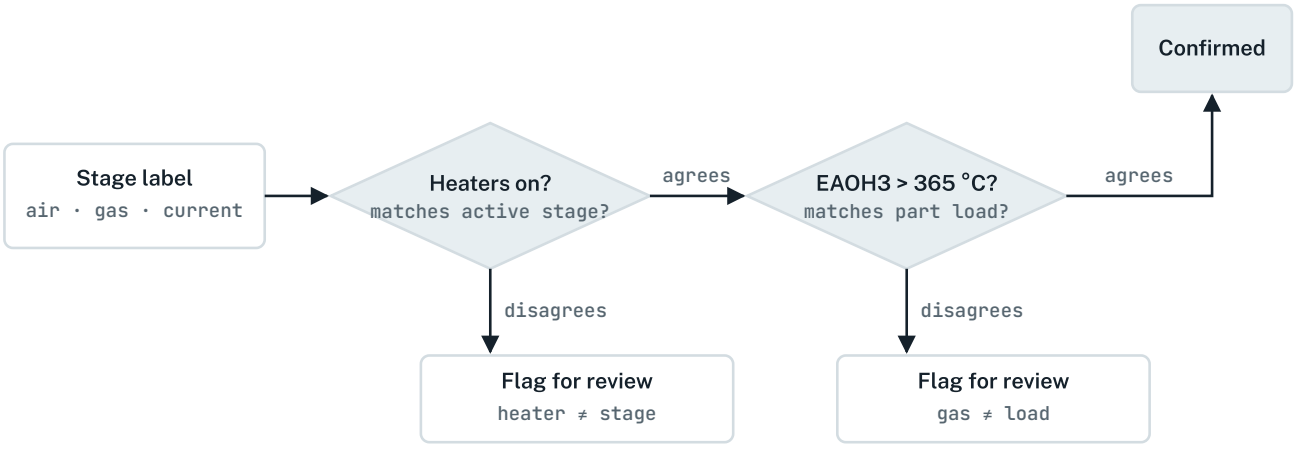
So EAOH3 above 365 °C says **"natural gas is flowing and power is likely"**. It is a good confirmation of part load, but on its own it also catches the minutes just before and after.

> **If you remember one thing:** EAOH3 temperature is the heater signal that tracks fuel and current. The others do not.

PAGE 4 · WHAT WE CAN DECIDE

# Use the heaters as a check, not as the judge

The decided stage rules (air flow, gas supply, current) should keep deciding the stage. The heaters add two cheap checks that catch wrong or doubtful labels.




Proposed cross-check. Nothing in today's stage rules changes; the two heater tests only confirm a label or flag it for a person to look at.

## Can the heaters name the stage by themselves?

We trained a small rule tree (a set of if-then questions, at most four deep) to name the stage from different signals, testing it each time on days it had not seen.




Share of minutes named correctly, all 39 day files, tested on unseen days.

| STAGE | STACK T ONLY | HEATERS ONLY | BOTH |
|---|---|---|---|
| Heat-up air | 63 % | 66 % | 58 % |
| Heat-up $N_2$+$H_2$ | 92 % | 79 % | 86 % |
| Hot standby | 88 % | 84 % | 92 % |
| Fuel switch | 30 % | 48 % | 40 % |
| Part load | 72 % | 83 % | 80 % |
| Air-flow cooling | 71 % | 74 % | 41 % |
| Shutdown | 85 % | 58 % | 74 % |
| Trip: purge | 71 % | 78 % | 51 % |

None of these comes close to the decided rules, which read the air, gas and current directly. The heaters alone do best on **Hot standby** (84 %) and **Part load** (83 %). Together with the stack they lift **Hot standby** from 88 % to 92 %, but they pull **Air-flow cooling** down from 71 % to 41 %: the heaters are off in both Air-flow cooling and Shutdown, and what separates those two is the air flow, which neither the heaters nor the stack temperature can see. **Fuel switch** stays hard for every signal set (30–48 %): it is short and is defined by natural gas flow, not by temperature.

## When do the heaters switch at a stage change?

| STAGE CHANGE | HEATERS SWITCH | TIMING |
|---|---|---|
| Shutdown / Air-flow cooling → Heat-up air | 10 of 12 (heaters come on) | ≈ 2 min after |
| Shutdown → Hot standby (re-heat after a stop) | 6 of 7 (heaters come on) | ≈ 2 min after |
| Hot standby → Fuel switch | 12 of 17 (some drop out) | ≈ 7 min after |
| Fuel switch → Part load | 7 of 10 (some drop out) | ≈ 9 min after |
| Hot standby → Air-flow cooling | 5 of 6 (heaters go off) | ≈ 13 min before |
| Part load → Air-flow cooling | 3 of 3 (heaters go off) | ≈ 6 min after |
| Any → Trip: purge | 6 of 6 (heaters go off) | same minute |
| Heat-up air → Heat-up $N_2$+$H_2$ → Hot standby | rarely | heaters stay on |

So a heater switch marks the start and end of the active period, the start of natural gas and the trips. It does not mark the steps between the heat-ups and Hot standby.

## Decisions for you

1. **Add the heater check** (heaters on ↔ active stage) to the stage checker? It agrees 93 % of the time; the rest are mostly the early part of Heat-up air, the known false starts and the first minutes of air cooling.
2. **Add EAOH3 > 365 °C** as a Fuel switch / Part load (natural gas) confirmation?
3. **Keep an eye on ERH:** its safety sensor peaked at 834 °C, 26 K under the 860 °C trip (13 Mar 2026). Should a warning level, for example 840 °C, be shown?
4. **Keep the decided stage rules as the only judge** (recommended), with heaters used only for checks?

Notebook: `data_analysis/heater_temperatures.ipynb` · analysis scripts kept in this session's scratch folder.

> **If you remember one thing:** keep the decided rules (air, gas, current) deciding the stage; let heaters on/off and EAOH3 temperature flag labels that look wrong.